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Uncertainty and restrictive migration rhetoric in parliament: evidence from Germany

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ABSTRACT

Whether political rhetoric becomes more reassuring or more restrictive during periods of uncertainty remains an open question in political economy. This paper examines the dynamic response of parliamentary migration discourse to uncertainty shocks. Using 28 years of German plenary debates (1991–2018), we construct a high-frequency Migration Sentiment Index (MSI) capturing rhetoric along an openness–restriction dimension. Local projection estimates show a systematic shift towards more restrictive rhetoric following increases in uncertainty, with consistently negative responses to migration-specific uncertainty and more uneven responses to broader uncertainty measures. Results are robust to controls for inflation, unemployment, and migration flows. These findings suggest that uncertainty is associated with more restrictive positioning by political actors in parliamentary debate.

KEYWORDS

Anti-immigration; Sentiment Index; political economy; parliamentary discourse; uncertainty

JEL CLASSIFICATION

C82; D72; F22

1. Introduction

How does parliament react when migration-related uncertainty rises? Theoretically, the direction of this relationship is ambiguous. On one hand, periods of volatility could prompt political actors to offer reassuring, ‘open’ narratives to stabilize public expectations and signal institutional competence. On the other hand, legislators may pivot towards restrictive signalling to align with perceived public anxiety and security concerns. This paper empirically examines which of these patterns characterizes German parliamentary migration discourse following uncertainty shocks.

The German Bundestag provides a useful setting for studying how migration rhetoric responds to different dimensions of uncertainty, as captured by established indices from the literature and described in `subsec:uncertainty_data`. Since reunification, German migration politics has moved through several distinct phases: asylum conflict from the early 1990s into the early 2000s, labour-market and integration reforms in the 2000s, and renewed politicization around the refugee crisis of 2015–2016. These episodes mirror broader global pressures linked to conflict, economic inequality,

displacement, and geopolitical instability, while also raising domestic questions about border control, labour-market adjustment, and integration capacity.

Existing research on migration sentiment typically relies on low-frequency data, such as party manifestos, which capture policy positions only at election-cycle intervals rather than during active legislative sessions (Lehmann et al. 2025). While survey-based measures and newspaper-indexed uncertainty provide valuable insights into public attitudes and media attention (Baker, Bloom, and Davis 2016; Ceylan and Ünlü 2024; Fraser and Ungor 2019; Wahlen and Mannheim 2026), they do not directly reflect the internal rhetorical climate of the legislature. What is missing is a systematic, high-frequency record of the migration tone within parliament itself. We fill this gap by shifting the analytical focus from slow-moving policy documents to the quarterly tenor of parliamentary discourse.

This letter makes two contributions to the literature. First, we construct a novel high-frequency Migration Sentiment Index (MSI) from the ParlSpeech V2 corpus (Rauh and Schwalbach 2020). The corpus covers 379,545 plenary speeches

delivered in the German Bundestag between 1991 and 2018 and contains 6,615,001 sentence units after sentence segmentation. After restricting the data to observations with the required metadata and applying the initial preprocessing filters, 2,318,207 sentences remain for the analysis. The resulting index permits a more granular analysis of political rhetoric than has previously been possible. Second, we document a systematic relationship in which heightened uncertainty is followed by a sustained shift towards more restrictive migration rhetoric.

Using a local projections framework, we find consistently negative responses following migration-specific uncertainty shocks, while responses to broader uncertainty measures fluctuate more across horizons. These results suggest that parliamentary discourse serves as a distinct transmission channel for uncertainty, where periods of volatility are characterized by a shift towards restrictive signalling rather than more open or reassuring narratives. Consistent with the descriptive nature of the analysis, these impulse responses are interpreted as informed correlations.

II. The migration sentiment index

Construction

The Migration Sentiment Index (MSI) is constructed from the ParlSpeech V2 corpus (Rauh and Schwalbach 2020), which contains 379,545 plenary speeches delivered in the German Bundestag between 1991Q1 and 2018Q4. Sentence segmentation yields 6,615,001 sentence units in the full corpus. After restricting the data to observations with the required metadata and applying the initial preprocessing filters, 2,318,207 sentences remain for the subsequent analysis. We retain sentences containing migration-related anchor terms, such as asylum, refugees, migration, deportation, residence status, nationality, or integration. This leaves 81,031 migration-related sentences, equivalent to 1.22% of the full segmented corpus.

For each anchor-selected sentence i , we assign two non-negative weights: an openness weight o_i and a restriction weight r_i . These weights are based on lexicon matches in the local context around

migration anchors. Openness weights capture language related to protection, rights, integration, legal access, family reunification, and inclusion. Restriction weights capture language related to deportation, border control, exclusion, limits, threat, and enforcement. Rule-based corrections account for negation and contextual critique, so that phrases such as ‘opposition to deportation’ are counted as openness-related rather than restriction-related language. Details are provided in Appendix A.

The index is aggregated through party-month log ratios. For party p in month m , define

$$O_{pm} = \sum_{i \in (p,m)} o_i, \quad R_{pm} = \sum_{i \in (p,m)} r_i, \\ T_{pm} = \log \left(\frac{O_{pm} + 0.5}{R_{pm} + 0.5} \right).$$

The constant 0.5 stabilizes ratios in party-months with few migration-related statements. The monthly MSI is the speech-volume-weighted average $MSI_m = \sum_p \omega_{pm} T_{pm}$, where ω_{pm} is party p ’s share of migration-related words in month m . The quarterly MSI is the average of monthly values within the quarter. Higher values indicate that openness-related language is more prevalent relative to restriction-related language.

Validation and empirical alignment

To establish the MSI’s external validity, we benchmark its trajectory against major German migration-policy episodes (Figure 1). The index successfully identifies key legislative shifts, reaching restrictive local minima during the 1993 Asylum Compromise (Asylkompromiss), which restricted constitutional asylum rights, and the 2016 Integration Act (Integrationsgesetz), which introduced stricter requirements for residency. Conversely, the series captures the expansionary rhetorical shift surrounding the 2005 Immigration Act (Zuwanderungsgesetz), a landmark reform that modernized Germany’s labour migration and residence framework.

Further validation confirms the index reliably maps to established party platforms. As detailed in Appendix A, the cross-sectional distribution of sentiment aligns with political priors: the Greens

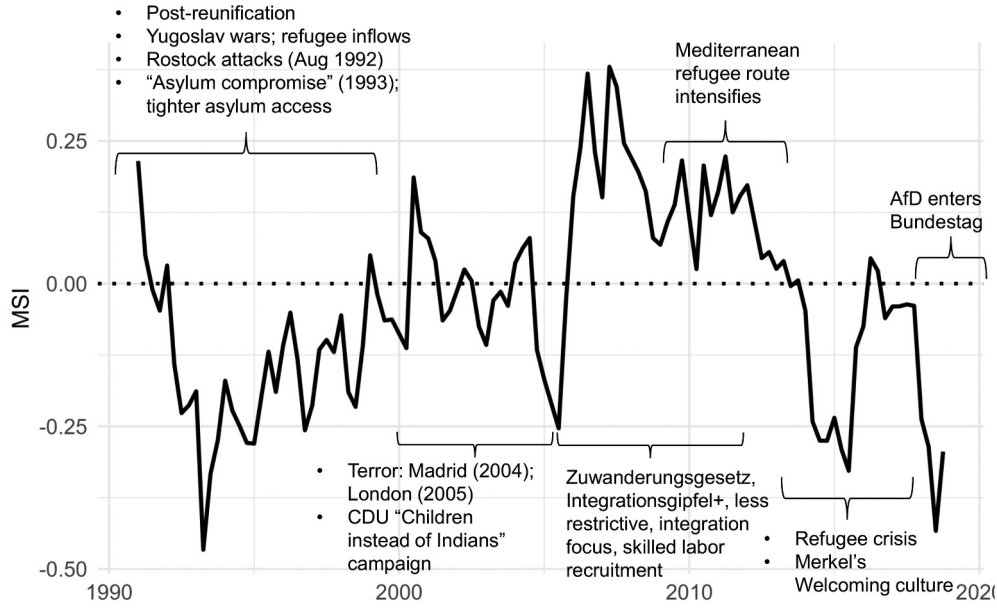


Figure 1. Historical evolution of parliamentary migration sentiment (MSI) and forensic validation. The figure plots an Exponentially Weighted Moving Average (EWMA) of the Migration Sentiment Index (MSI). Values above (below) zero denote a parliamentary tone that is more open (restrictive) relative to the 1991–2018 average.

consistently occupy the upper (open) tail, while the Alternative for Germany (AfD) occupies the extreme lower (restrictive) tail.

III. Empirical strategy

Data

We assemble quarterly data from established uncertainty measures and standard macroeconomic indicators to examine the drivers of parliamentary migration tone. Data sources are presented in Appendix Table C.1.

Our primary independent variables are four established text-based uncertainty indices for Germany, selected to capture both migration-specific uncertainty and broader economic or geopolitical uncertainty. Following the text-based uncertainty literature, we interpret these indices as proxies for reduced clarity about future conditions, policy decisions, the timing and content of policy actions, and their consequences (Baker, Bloom, and Davis 2016). In the migration context, this includes uncertainty about future migration policy, migration-related pressures, and the economic and administrative consequences of policy action or inaction. The four indices are the Germany-specific versions of the Migration Fear

Index, the Migration Policy Uncertainty Index, the Economic Policy Uncertainty Index, and the World Uncertainty Index (WUI). While the EPU index captures uncertainty about economic policy, the WUI captures broader global uncertainty, including geopolitical and international risks. Monthly series are converted to quarterly frequency via arithmetic means to align with the MSI.

To isolate the association between uncertainty and migration sentiment from broader cyclical fluctuations, we include a compact control for macroeconomic fundamentals. Given the limited quarterly sample, we summarize unemployment, inflation, and output growth in a composite Macroeconomic Stress Index:

$$\text{Stress}_t = z(\Delta u_t) + z(\pi_t) - z(g_t),$$

where Δu_t , π_t , and g_t denote the change in the unemployment rate, the inflation rate, and real GDP growth, respectively. The function $z(\cdot)$ denotes standardization over the estimation sample:

$$z(x_t) = \frac{x_t - \bar{x}}{\sigma_x},$$

where $\bar{x}$ and σ_x are the sample mean and standard deviation of x_t . Higher values of Stress_t therefore correspond to rising unemployment, higher

inflation, and weaker real GDP growth. For extended robustness specifications, we further augment the model with net migration flows from the German Federal Statistical Office (*Statistisches Bundesamt*) and survey-based public concern regarding migration from *Politbarometer* (*Statistisches Bundesamt Destatis* 2026; Wahlen and Mannheim 2026). All variables are standardized prior to estimation to ensure comparability across specifications.

Local projections

We estimate the response of parliamentary migration sentiment to uncertainty shocks using local projections following Jordà (2005). To preserve degrees of freedom in the short quarterly sample, we use a parsimonious bivariate specification in which uncertainty and the Migration Sentiment Index (MSI) form the endogenous block, while macroeconomic stress and other robustness variables enter as exogenous controls. All variables are standardized prior to estimation.

For each horizon $h = 0, \dots, 4$, we estimate

$$\begin{aligned} MSI_{t+h} = & \alpha_h + \beta_h U_t + \sum_{j=1}^p \Gamma_{h,j} U_{t-j} \\ & + \sum_{j=1}^p \Phi_{h,j} MSI_{t-j} + \sum_{j=0}^p \Lambda_{h,j} X_{t-j} + \varepsilon_{t+h}, \end{aligned}$$

where U_t denotes one uncertainty measure at a time, and X_t includes the macroeconomic stress index and a refugee-crisis dummy. In extended-sample robustness specifications, X_t additionally includes realized net migration flows and survey-based public concern about migration from *Politbarometer*.

Uncertainty shocks are identified recursively by ordering uncertainty before the MSI. This ordering reflects the idea that parliamentary migration tone can respond within the same quarter to changes in uncertainty, since legislators can quickly adjust their rhetoric to the broader political and economic environment. By contrast, any effect of parliamentary tone on uncertainty is more plausibly delayed, as it would likely operate through subsequent

policy action or more persistent shifts in political discourse rather than through a one-quarter rhetorical shock.

We use one lag, following the BIC selection criterion, report horizons up to four quarters, and compute Newey-West standard errors. The figures report nested 68, 90, and 95% confidence intervals. The baseline sample covers 1996Q1–2018Q4, while robustness specifications using migration flows and *Politbarometer* are estimated over 2008Q1–2018Q4. As in Azzimonti (2018), the resulting impulse responses are interpreted as informed correlations rather than causal effects.

IV. Results

Figure 2 reports the response of the Migration Sentiment Index (MSI) to uncertainty shocks. The responses to both migration fear and migration policy uncertainty are negative throughout the horizon, indicating a shift towards more restrictive parliamentary rhetoric. The response to migration fear becomes progressively more negative, whereas the response to migration policy uncertainty is less monotonic. The responses to the broader uncertainty measures vary more across horizons. The Germany-specific Economic Policy Uncertainty index produces a negative response on impact and again at later horizons, while the response to the World Uncertainty Index for Germany is negative overall but fluctuates considerably.

In Appendix B, we report a battery of robustness checks (Figures B.2 through B.5). The negative response remains visible when controlling for macroeconomic stress and realized net migration flows. When survey-based public concern about migration is added in the shorter extended sample, the response becomes weaker, but the medium-horizon pattern remains negative.

V. Conclusion

This paper constructs a new quarterly measure of parliamentary migration sentiment for Germany from Bundestag plenary debates. Using local projections, it shows that increases in uncertainty are

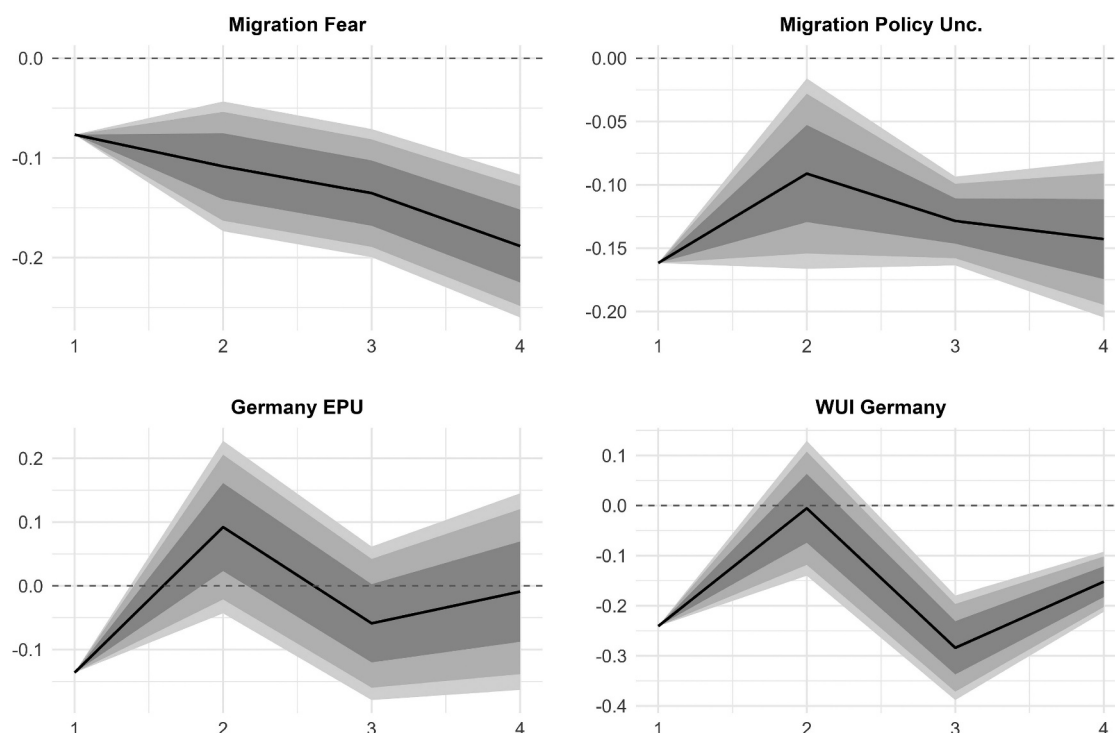


Figure 2. Response of the migration sentiment Index (MSI) to uncertainty shocks. *Notes:* The figure reports local projection impulse responses of the Migration Sentiment Index (MSI) to shocks in four uncertainty measures. Higher MSI values indicate a more open tone; negative responses denote a shift towards restrictive rhetoric. The horizontal axis denotes horizons in quarters. The solid line shows the point estimate. Shaded areas show nested 68, 90, and 95% confidence intervals, from darkest to lightest. Standard errors are Newey-West corrected.

followed by a more restrictive parliamentary migration tone, with the strongest responses appearing for migration-specific uncertainty measures. In that sense, the evidence points more towards a restrictive than a reassuring parliamentary response when migration-related uncertainty rises.

The paper therefore contributes both a new high-frequency measure of legislative migration discourse and evidence of a systematic negative relationship between uncertainty and parliamentary migration sentiment. These results are descriptive rather than causal, and future work should examine more closely the policy and macroeconomic consequences of shifts in parliamentary migration tone.

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Author contributions

CRedit: **Youssuf Abdelatif:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing; **Zo Andriantomanga:** Conceptualization, Formal analysis, Methodology, Supervision, Visualization, Writing – original draft, Writing – review & editing.

Disclosure statement

No potential conflict of interest was reported by the author(s).

ChatGPT (OpenAI) was used for linguistic and code refinement. All outputs were reviewed and verified by the authors.

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Data availability statement

The authors confirm that the data supporting the findings of this study are publicly available.

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